SHRI MADHWA VADIRAJA INSTITUTE OF TECHNOLOGY AND MANAGEMENT
An Autonomous Institute Affiliated to VTU, Belagavi
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Vishwothama Nagar, Bantakal – 574115, Udupi District, Karnataka
SMVITM

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT PROPOSAL

PROJECT GROUP NO: **ACADEMIC YEAR: 2026-27**

A. Proposed Title of the Project

AI-Based Drowning and Distress Detection in Swimming Pools.

B. Introduction

Drowning is a major public health problem: the WHO ranks it as the third most frequent cause of unintentional injury death, with roughly 236,000 deaths a year worldwide [2]. Pools are supposed to be supervised environments, yet a drowning swimmer rarely thrashes or calls for help. The struggle is quiet, short and easily hidden among other swimmers, so a lifeguard watching a crowded pool can miss it.

Camera-based AI offers a continuous second pair of eyes. Reviews of the field find that vision-based deep learning now dominates drowning-detection research, with YOLO-family detectors the most common choice [1], [2], [3]. Many published systems, however, concentrate on detecting the drowning state in individual frames. Following each swimmer over time, so that a brief dive can be told apart from a swimmer who has stopped moving, and linking detection to staff alerts, receive less attention.

This project builds an end-to-end prototype for swimming pools. A YOLO detector finds swimmers in each frame, a multi-object tracker gives every swimmer a persistent ID, a temporal behaviour module watches each track for distress patterns, and a risk score decides when to raise an alert on a live dashboard. The aim is early, low-false-alarm warning that supports lifeguards rather than replacing them.

C. Literature Review

Seventeen papers were reviewed (numbering follows Section L). They fall into four groups.

1. **Surveys of drowning detection.** Jalalifar et al. [1] review image-processing and sensor-based approaches and conclude these are the most effective families of drowning detection. Shatnawi et al. [2] split systems into detection-only and detection-plus-rescue, and into computer-vision and sensor-based methods. They call for stronger AI models, uniform datasets and tighter links between detection and rescue. Alharbi [3] finds that machine learning dominates raw sensor analysis while deep learning drives vision, notes a gap between advanced vision models and practical detection, and stresses real-time, low-latency operation. Alotaibi [13] combined IoT with transfer learning for pool monitoring; [5] points out that such models may generalise poorly across varied drowning situations.

2. **YOLO-based pool detectors.** Zhang et al. [4] propose an improved YOLO11 model with attention and feature fusion, aimed at lightweight real-time deployment in pools. Song et al. [5] present MS-YOLO, built on YOLOv8, which reaches 86.4% average accuracy on a self-built dataset at only 7.3 GFLOPs, and note that public drowning datasets are scarce. Zeng et al. [6] introduce DrownACB-YOLO, using atrous spatial pyramid pooling and CARAFE up-sampling; because most methods only recognise the drowned state, they add a transition label to give earlier alerts. Hong et al. [9] frame a two-stage pipeline (swimmer detection, then risk-behaviour classification) and benchmark YOLOv12 against RT-DETR for the first stage, though their experiments cover only that stage.

3. **Other detectors and behaviour recognition.** Hayat et al. [8] use Mask R-CNN to detect near-drowning

persons in pools. Lei et al. [7] recognise drowning behaviour from underwater video sequences with deep learning. Reference [10] applies YOLOv5 and Faster R-CNN to infant drowning, where small body size and limited motion make detection harder. Reference [11] targets coastal waters, segmenting the sea region and then combining YOLO with a Vision Transformer over consecutive frames to spot anomalies. Claesson et al. [12] tested drones with machine learning to recognise simulated drowning victims in open water.

4. **Detection and tracking foundations.** YOLO [14] introduced single-pass real-time detection. SORT [15] pairs a Kalman filter with Hungarian matching for fast tracking, DeepSORT [16] adds appearance features to keep IDs through crossings, and ByteTrack [17] keeps low-confidence detections to recover blurred or partly hidden targets, which suits splashes and occlusion in pools.

   Research gap. Many pool detectors focus on recognising the drowned state in single frames or short clips [6]. Reviews call for uniform datasets and complete detection-to-alert pipelines [2], [3], and [9] leaves behaviour classification as future work. This project addresses that gap with a tracking-based pipeline that scores each swimmer's behaviour over time and raises an alert only when the risk persists.

D. Objective

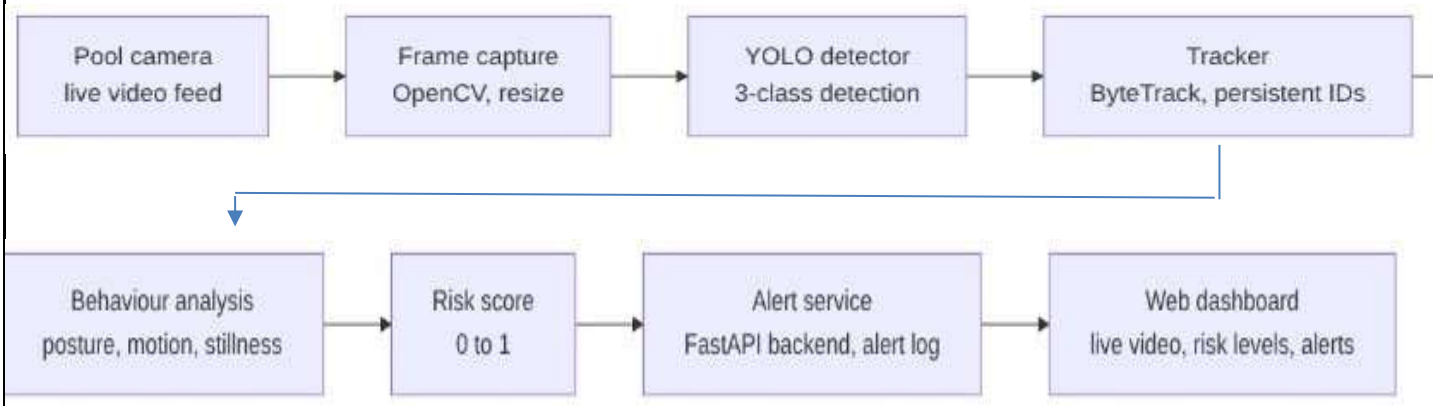
Main aim: to design and evaluate a real-time, camera-based system that detects drowning and distress behaviour of swimmers in a pool and alerts supervisors early.

E. Workplace	Address of Other Institute / Industry (if any):
Shri Madhwa Vadiraja Institute of Technology and Management (SMVITM), Department of CSE	Vishwothama Nagar, Bantakal, Udupi District, Karnataka 574115 (work is done at the college; no external institute or industry involved)

F. Requirement Details

1. Hardware	Laptop/PC with an NVIDIA GPU (8 GB VRAM or more recommended) for training; 16 GB RAM.
2. Software	Python, PyTorch, Ultralytics YOLO, OpenCV, NumPy, Pandas, Matplotlib, FastAPI, a web dashboard (HTML/JavaScript), Google Colab or Kaggle for training, Git/GitHub
3. Skill sets	Deep learning and computer vision, object detection and tracking, Python programming, REST API and web development, dataset annotation, model evaluation
4. Others	Public "Swimming & Drowning Detection" dataset (12,365 images; classes: Swimming, Drowning, Person out of water); pool or stock videos for testing; permission from the college if any pool footage is recorded

G. Figure(s)/Drawing(s)

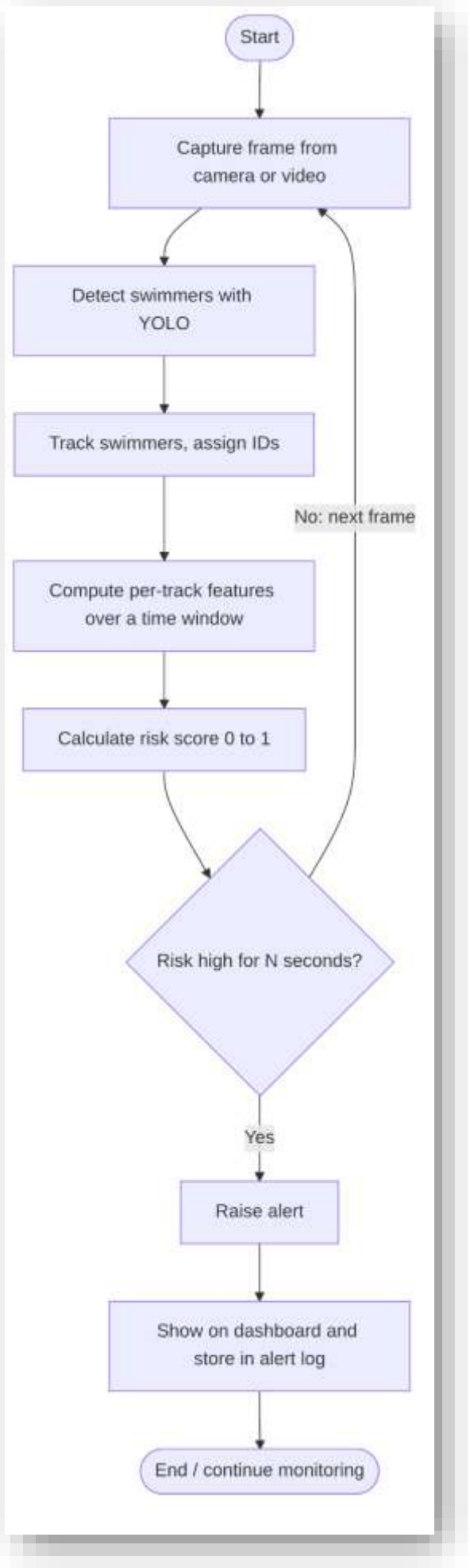


H. Methodology

The work follows seven stages.

1. **Data preparation**. Inspect the public Swimming & Drowning Detection dataset (12,365 images, three classes), check class balance, split into train, validation and test sets, and apply augmentation (brightness, blur, flip, scale) to mimic glare, ripples and camera angles. Collect a few pool or stock videos for testing on unseen footage.
2. **Detection.** Fine-tune an Ultralytics YOLO model (a small variant such as YOLOv8n or YOLO11n first, a larger one for comparison). Measure precision, recall and mAP, and record the speed in frames per second.
3. **Tracking**. Run the detector on video and attach a tracker (ByteTrack first, DeepSORT or BoT-SORT for comparison) so every swimmer has a persistent ID. Compare trackers on ID switches and stability when swimmers cross or are partly hidden.
4. **Temporal behaviour analysis**. For each track, keep a sliding window of recent frames and compute features such as the drowning-class confidence, bounding-box height-to-width ratio (an upright posture), vertical position relative to the waterline, motion between frames, and how long the swimmer has stayed nearly still.
5. **Risk scoring and alerts**. Combine the features into a weighted risk score from 0 to 1 with Low, Medium and High levels. Raise an alert only when the score stays above a threshold for a set number of seconds, which reduces false alarms from a single odd frame. Thresholds are tuned on validation videos.
6. **Dashboard**. A FastAPI backend streams annotated video, track IDs, risk levels and alert events to a web dashboard, and stores an alert log for review.
7. **Testing and evaluation**. Test under difficult conditions (surface glare, splashing, crowded lanes, occlusion, outdoor lighting). Report detection metrics, alert latency, false-alarm rate per hour of video, missed events and FPS, then compare design choices and write the final report.

I. Flowchart (Plan of Action)



J. Expected Outcome of the project

- A trained YOLO detector for swimming, drowning and person-out-of-water classes, with a reported mAP, precision and recall on a held-out test set.
- A working tracking and behaviour-analysis pipeline that assigns each swimmer a risk score in real time.
- A web dashboard that shows live video, risk levels and a log of alerts.
- A comparison of detector and tracker choices, with measured alert delay, false-alarm rate and FPS under hard conditions.
- A project report, a demo video and a research-style write-up, with the prototype positioned as an aid to lifeguards, not a replacement.

K. Estimated cost	**Source of Fund**
The software stack is free and open source, and training uses the free Colab/Kaggle sessions, so the base cost is close to zero.	self-funded by the project team (no external sponsor)

L. References

Found through web search; items marked † had incomplete details in the sources opened and must be completed from the paper itself before submission.

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2. M. Shatnawi, F. Albreiki, A. Alkhoori, M. Alhebshi, and A. Shatnawi, "Advances and challenges in automated drowning detection and prevention systems," *Information*, vol. 15, no. 11, p. 721, 2024, doi: 10.3390/info15110721.
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10. † "Automatic real-time detection of infant drowning using YOLOv5 and Faster R-CNN models based on video surveillance," IEEE Xplore, document 10184087 (authors, journal and year to be added).
11. † "Drowning detection algorithm in coastal zones," in *Proc. 7th Int. Conf. Inf., Control, Commun. Technol. (ICCT)*, IEEE, 2023, IEEE Xplore document 10347125 (authors to be added).
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13. † A. Alotaibi, "Automated and intelligent system for monitoring swimming pool safety based on the IoT and transfer learning," 2020 (journal and pages to be added).
14. J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 779–788.
15. A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, 2016, pp. 3464–3468.
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M. Project Batch members

S.No.	USN	Name	Signature with date
1.	4MW24CS119	Sagar H Poojary	
2.	4MW24CS117	Rohan Mendon	
3.	4MW24CS090	Nikhil Naik	
4.	4MW24CS076	Manoj Kumar	

N. Accepted/Rejected:	

O. Suggestions for implementation:

P. Guidance

Guide (s) allotted	Name and Signature (s) with date

1. **Guide** 2. **Reviewer** 3. **Reviewer**	

Q. Review Comments/Suggestions **Accept/Modify/Reject**